

# Pre-requisites

## 1. Overview

The EduGenie project is an AI-driven learning portal designed to assist users with multiple learning tasks, including concept explanation, general Q&A, content summarization, quiz generation, and learning path creation. To run, build, and deploy this project successfully, specific software tools, programming libraries, and API credentials must be properly configured. This document details the complete environment setup required to initialize the project ecosystem.

## 2. Development Tools and Environment

**Python Runtime:** The core application logic is built using Python. Python version 3.10 or higher is required to guarantee compatibility with asynchronous processing patterns and modern deep learning packages.

**Integrated Development Environment (IDE):** Visual Studio Code (VS Code) or PyCharm is recommended for writing code, debugging FastAPI paths, and managing configuration (.env) frameworks.

**System Requirements:** A modern CPU is mandatory. Due to the local execution of a Seq2Seq transformers model, a minimum of 8GB RAM is required to hold model weights in memory during runtime generation.

## 3. Core Libraries and Dependencies

The project uses standard configuration dependencies and specialized AI frameworks managed via a requirements.txt manifest:

**Web Framework:** 'fastapi' and 'uvicorn' manage high-performance routing and serve as the production-ready Asynchronous Server Gateway Interface (ASGI) server.

**Templating Engines:** 'jinja2' renders responsive HTML views dynamically on the server side.

**AI Orchestration & API Clients:** 'google-genai' serves as the programmatic SDK interface to route text logic directly to cloud-hosted intelligence models.

**Local Machine Learning Ecosystem:** 'transformers', 'torch', 'sentencepiece', and 'protobuf' assemble the local deep learning stack, allowing tokenization and generation loops to execute directly on the host CPU.

## 4. API Keys and Environment Management

Secure storage of sensitive parameters is handled via a localized .env variables system.

**Gemini API Access:** The developer must secure an API key from Google AI Studio. This project expressly prohibits standard OAuth user tokens (identifiable by a ya29. prefix) because the Google GenAI client expects dedicated, backend-authorized API credentials.

**Environment File:** A .env document must be created inside the project root folder declaring GEMINI\_API\_KEY or GOOGLE\_API\_KEY to prevent application initialization failures during startup verification.